

Abstract

We study the topological properties of phonons in trapped Rydberg atom arrays, which arise from dipole-dipole interactions between the atoms. For various one-dimensional geometries, from zigzag to armchair configurations, we analyze the symmetries of the phononic Hamiltonian which give rise to topologically localized phonon excitations on the system boundaries. Because the phonon-phonon interactions here do not naturally respect chiral symmetry, which is crucial for realizing 1D topological phases, we show how an appropriate, geometry-dependent choice of the dipole moment orientation can restore this symmetry and enable topological characterization. The interplay between two phononic degrees of freedom in the harmonic trap potential, mediated by the dipole-dipole interactions, leads to interesting topological phases with winding numbers between zero and two. The corresponding edge states exhibit strong localization and robustness, making them potentially useful for applications in Rydberg-array-based quantum transport.

As an outlook we aim to explore how the phonon topology influences transport phenomena. Topological states in the electronic degrees of freedom in Rydberg arrays have already been shown to support topologically protected photon pumping [1–5]. Studying the effect of spin-phonon coupling can provide new control mechanisms in the realm of topological transport.

Hamiltonian

I am considering the interaction term H_I of a Hamiltonian of a system of N particles with positions $\{\bar{\mathbf{R}}_l\}_{l \in \{1, \dots, N\}}$. It shall consist of two-body interactions, which depend on the position of the involved particles. Thus H_I can be written in the form

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j; i \neq j} V_{i,j}(\bar{\mathbf{R}}_i, \bar{\mathbf{R}}_j)$$

In order to being able to derive the second order expansion of H_I , that will be needed later, in a more formal way, I write the two-body interactions as dependend on all positions.

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j; i \neq j} V_{i,j}(\{\bar{\mathbf{R}}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\bar{\mathbf{R}}_l\}$, i.e. $V_{i,j}(\{\bar{\mathbf{R}}_l\})$. But of course they only depend on $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{R}}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \bar{\mathbf{R}}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the interaction Hamiltonian in small displacements $\{\delta \mathbf{R}_l\}$ from the tweezer centers $\{\mathbf{R}_l\}$.

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n,m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m + \frac{1}{2} \sum_{n=m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &\quad + \frac{1}{2} \sum_n \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m (\delta_{n=i, m=j} + \delta_{n=j, m=i}) \\ &\quad + \frac{1}{2} \sum_n \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n (\delta_{n=i} + \delta_{n=j}) \end{aligned}$$

Using the fact, that

$$\delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j = \delta \mathbf{R}_j \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \delta \mathbf{R}_i$$

the expansion can be written as

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \delta \mathbf{R}_i + \sum_{i,j; i \neq j} \delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j \\ &+ \sum_{i,j; i \neq j} \delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i \end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \sum_{n,m} \delta \mathbf{R}_n U_{n,m} \delta \mathbf{R}_m$$

with

$$\begin{aligned} U_{n,m} &= \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \end{aligned}$$

Swithing again to the reduced writting, where V_{ij} is only written as a function of the variables of which it actually depends, reads

$$\begin{aligned} U_{n,m} &= \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial \mathbf{R}_n^2} \end{aligned}$$

This result is a good point to further specify the potential that is used in the next sections. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_i} = \mathbb{1} = -\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} &= \frac{\partial}{\partial \mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \right) \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial^2 \mathbf{r}_{nm}}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \\ &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_m^2} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \end{aligned}$$

Thus

$$\begin{aligned} U_{n,m} &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2} \end{aligned}$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$H_0(\{\delta\mathbf{R}_l\}) = \sum_n \frac{1}{2} m\omega^2 \delta\mathbf{R}_n^2$$

and redefining the trap centers by a equal shift of all positions $\{\delta\mathbf{R}_l\} = \{\delta\tilde{\mathbf{R}}_l + \mathbf{\Delta}\}$. Neglection all constant terms the Hamiltonian up to second order reads

$$\begin{aligned} H(\{\delta\mathbf{R}_l\}) &= \sum_n \frac{1}{2} m\omega^2 (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n)^2 + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n) + \sum_{n,m} (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n) U_{n,m} (\delta\tilde{\mathbf{R}}_m + \mathbf{\Delta}_m) \\ &= \sum_n \frac{1}{2} m\omega^2 \delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta\tilde{\mathbf{R}}_n U_{n,m} \delta\tilde{\mathbf{R}}_m + \sum_n \left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} + \left(2 \sum_m \delta\tilde{\mathbf{R}}_m U_{m,n} + m\omega^2 \right) \mathbf{\Delta}_n \right) \delta\tilde{\mathbf{R}}_n, \end{aligned}$$

where it was used that U is symmetric. The linear term vanishes, if $\mathbf{\Delta}_n$ is chosen such that

$$\mathbf{\Delta}_n = - \left(2 \sum_m \delta\tilde{\mathbf{R}}_m U_{m,n} + m\omega^2 \right)^{-1} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}$$

Thus

$$H(\{\delta\tilde{\mathbf{R}}_l\}) = \sum_n \frac{1}{2} m\omega^2 \delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta\tilde{\mathbf{R}}_n U_{n,m} \delta\tilde{\mathbf{R}}_m$$

To describe the deviation from the center in a quantum mechanical way, the displacements are now quantized as phonons using the quadrature operators

$$\delta\tilde{\mathbf{R}}_n = \frac{1}{\sqrt{m\omega}} (a_n + a_n^\dagger)$$

If we exploit that the interactions realized in U are much smaller compared to the trap frequency, i.e. $U_{n,m} \ll m\omega^2$, the creation and annihilation of phonons is suppressed and the dynamics only involves phonon hopping, thus combinations aa or $a^\dagger a^\dagger$ are discarded.

$$\begin{aligned} H &= \sum_n \omega a_n^\dagger a_n + 2 \sum_{n,m} \frac{U_{n,m}}{m\omega} a_n^\dagger a_m \\ &=: \sum_n \omega a_n^\dagger a_n + \sum_{n,m} \tilde{U}_{n,m} a_n^\dagger a_m \end{aligned}$$

Momentum space Hamiltonian

Let $c_{\mathbf{R},\alpha}^A$ and $c_{\mathbf{R},\alpha}^B$ be the phonon annihilation operators of sublattice A and B respectively at the unit cell, located at $\mathbf{R}$, where $\alpha \in \{x, y\}$ labels the the direction of movement. Let further L be the set of all unit cells of the lattice, and $C = \{c_{\mathbf{R},\alpha}^A, c_{\mathbf{R},\alpha}^B\}_{\mathbf{R} \in L, \alpha \in \{x, y\}}$ the set of all annihilation operators. The Hamiltonian in real space can then be written as

$$H = \sum_{\substack{\mathbf{R}, \mathbf{R}' \in L \\ a, b \in C}} H^{a,b}(\mathbf{R} - \mathbf{R}') a_{\mathbf{R}}^\dagger b_{\mathbf{R}'}$$

For two dimensional periodic boundary conditions one can introduce the Fourier transformed operators, $a \in C$

$$a_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{R} \in L} e^{-i\mathbf{k} \cdot \mathbf{R}} a_{\mathbf{R}},$$

$k \in \bar{L}$ where $\bar{L}$ is the reciprocal lattice of L and N is the extent of the lattice in one direction. Writing the Hamiltonian in terms of these operators yields

$$\begin{aligned}
H &= \frac{1}{N^2} \sum_{\substack{\mathbf{R}, \mathbf{R}' \in L \\ a, b \in C \\ \mathbf{k}, \mathbf{k}' \in \bar{L}}} H^{a,b}(\mathbf{R} - \mathbf{R}') e^{-i\mathbf{k}\mathbf{R}} e^{i\mathbf{k}'\mathbf{R}'} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'} \\
&= \frac{1}{N^2} \sum_{\substack{\mathbf{R}, \Delta\mathbf{R}=\mathbf{R}-\mathbf{R}' \\ a, b \\ \mathbf{k}, \mathbf{k}'}} H^{a,b}(\Delta\mathbf{R}) e^{-i\mathbf{R}(\mathbf{k}-\mathbf{k}')} e^{-i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'} \\
&= \sum_{\substack{\Delta\mathbf{R}, \mathbf{k} \\ a, b}} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{R}\mathbf{k}} a_{\mathbf{k}}^\dagger b_{\mathbf{k}} =: \sum_{\substack{\mathbf{k} \\ a, b}} H_{\mathbf{k}}^{a,b} a_{\mathbf{k}}^\dagger b_{\mathbf{k}} \\
\text{where } H_{\mathbf{k}}^{a,b} &= \sum_{\Delta\mathbf{R}} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{R}\mathbf{k}}
\end{aligned}$$

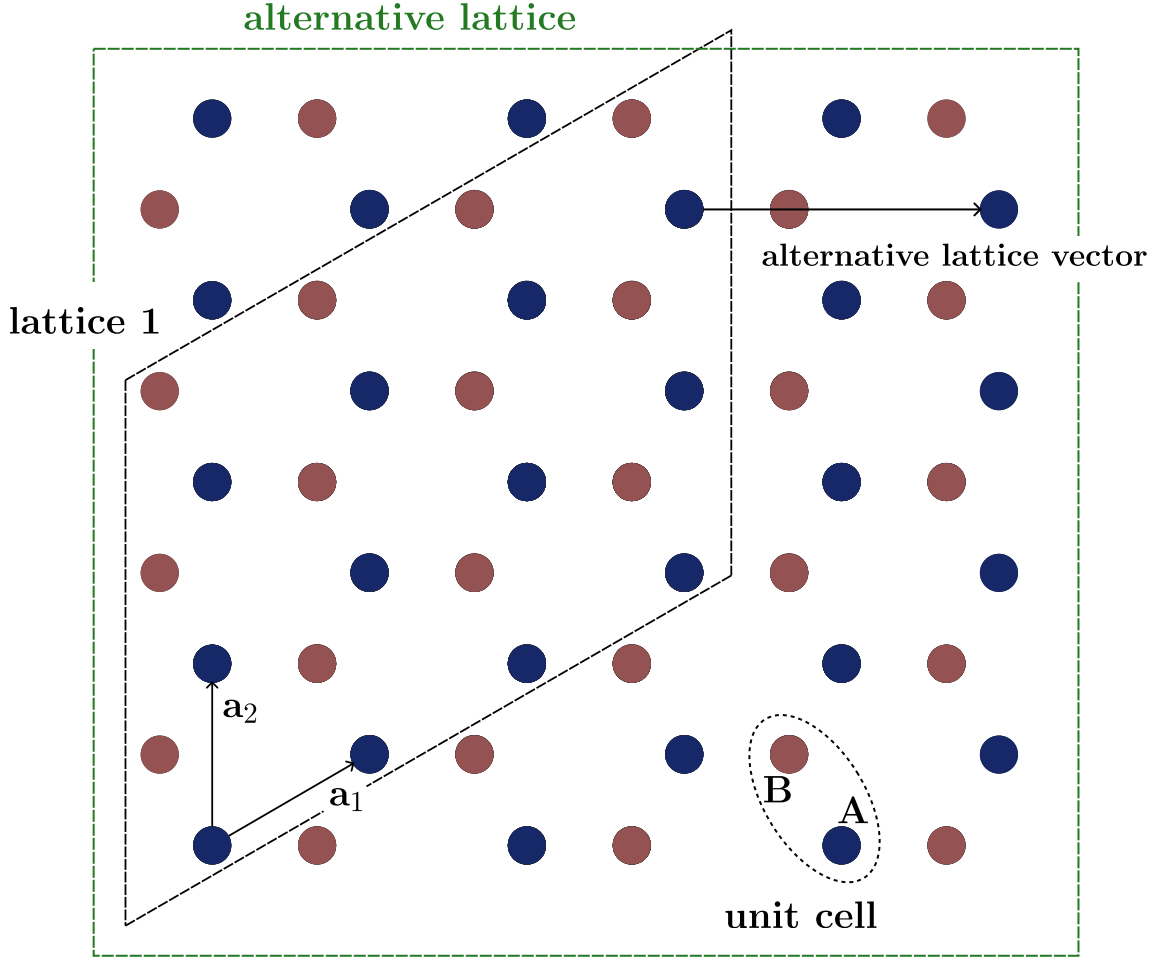


Figure 1: **Lattice definition** The main configuration of the grid, lattice 1, is the parallelogram spanned by the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. This is visualized by the black dashed frame. There is also the possibility of defining a square lattice which is highlighted by the green dashed frame.

For the case where only one direction of the lattice obeys periodic boundary conditions, while the other direction has open boundaries, the Fourier transform is only performed in the periodic direction. For this purpose I introduce the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$, which span the lattice. The lattice can be split into subspaces $L_n = \{j \cdot \mathbf{a}_n\}_{j \in M_n}$ with $n \in \{1, 2\}$ and M_n is chosen in such a way, that L_n spans from

one edge of the lattice to the other. Hence the position of any unit cell $\mathbf{R}$ can be expressed as the linear combination of two vectors $\mathbf{r}_1 \in L_1$ and $\mathbf{r}_2 \in L_2$, i.e. $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2$. I can now also introduce the reciprocal lattice vectors $\mathbf{k}_1$ and $\mathbf{k}_2$ with the property $\mathbf{a}_i \cdot \mathbf{k}_j = 2\pi\delta_{ij}$. I perform the Fourier transformation as an example for the lattice 1 from Fig. 1.

Here the lattice direction represented by $\mathbf{a}_2$ shall be periodic. I introduce the one dimensionally Fourier transformed operators

$$a_{\mathbf{r}_1, \mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_2 \in L_2} e^{-i\mathbf{k}\mathbf{r}_2} a_{\mathbf{r}_1 + \mathbf{r}_2}, \quad \mathbf{k} \in \left\{ \frac{1}{N}, \dots, 1 \right\} \cdot \mathbf{k}_2$$

The Hamiltonian can now be transformed via

$$\begin{aligned} H &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \in L_1 \\ \mathbf{r}_2, \mathbf{r}'_2 \in L_2}} \sum_{\substack{a, b \in C \\ \mathbf{k}, \mathbf{k}' \in \bar{L}}} H^{a,b}(\mathbf{R} - \mathbf{R}') e^{-i\mathbf{k}\mathbf{r}_2} e^{i\mathbf{k}'\mathbf{r}'_2} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}'} \\ &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \mathbf{r}_2, \Delta\mathbf{r}_2 = \mathbf{r}_2 - \mathbf{r}'_2}} \sum_{\substack{a, b \\ \mathbf{k}, \mathbf{k}'}} H^{a,b}(\Delta\mathbf{R}) e^{-i\mathbf{R}(\mathbf{k} - \mathbf{k}')} e^{-i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}'} \\ &= \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \Delta\mathbf{r}_2, \mathbf{k} \\ a, b}} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{r}_2\mathbf{k}} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}} =: \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \mathbf{k}, a, b}} H^{a,b}(\mathbf{k}, \Delta\mathbf{R}) a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}} \\ \text{where } H^{a,b}(\mathbf{k}, \Delta\mathbf{R}) &= \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{r}_2\mathbf{k}} = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{R}\mathbf{k}} \end{aligned}$$

The last equality holds, when $\mathbf{k}$ is in the direction of one of the reciprocal lattice vectors. However, as we perform only a one dimensional Fourier transform, I think a restriction to a purely one dimensional treatment is necessary. Thus $\mathbf{k}_n$ should only be chosen with respect to the condition $\mathbf{k}_n \mathbf{a}_n = 2\pi$, which gives the natural choice $\mathbf{k}_n \parallel \mathbf{a}_n$. For the case of the alternative square lattice I think, we should consider the number of rows twice as big as the number of columns and choose an alternative lattice vector into the x-direction, as depicted in Fig. 1.

What the one dimensional Fourier transform does, is projecting the system onto the one-dimensional chain not of atoms, but of interacting rows or columns of atoms. In this sense the division into sub-directions or sub-bases does not have to be restricted to the lattice vectors. The only thing needed is a labeling of the different rows or columns with the complementary direction having a periodic structure. This allows the division of the alternative square lattice as described above.

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subscripts labeling the different positions are dropped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = \frac{1}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^5},$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j ($i \neq j$). The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3r_\alpha}{r^5} + \frac{15r_\alpha(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - \frac{6(\mathbf{r} \cdot \hat{\mathbf{d}})d_\alpha}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta}}{r^5} + 15 \frac{r_\alpha r_\beta}{r^7} + 15 \frac{\delta_{\alpha\beta} (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \\
&\quad + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) - 6 \frac{d_\alpha d_\beta}{r^5} \\
&= \delta_{\alpha\beta} \left(-3 \frac{1}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} \right) - 6 \frac{d_\alpha d_\beta}{r^5} + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) \\
&\quad + 15 r_\alpha r_\beta \left(\frac{1}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \right).
\end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{r}) = \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5}.$$

The first derivative reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2) r_\alpha}{r^5} + \frac{15 r_\alpha (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - \frac{3 \left((\mathbf{r} \cdot \hat{\mathbf{d}}_1) d_{2,\alpha} + (\mathbf{r} \cdot \hat{\mathbf{d}}_2) d_{1,\alpha} \right)}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta} (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^5} + 15 \frac{r_\alpha r_\beta (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^7} + 15 \frac{\delta_{\alpha\beta} (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&= \delta_{\alpha\beta} \left(-3 \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} \right) - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] \\
&\quad + 15 r_\alpha r_\beta \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \right).
\end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned}
\frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2 V_{dd} \frac{\partial V_{dd}}{\partial r_\alpha} \\
\frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2 \frac{\partial V_{dd}}{\partial r_\alpha} \frac{\partial V_{dd}}{\partial r_\beta} + 2 V_{dd} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta}
\end{aligned}$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \quad (0.1)$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \quad (0.2)$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (0.3)$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy -plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy -plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \rightarrow \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z} dz = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [6, 7]

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr} [h^{-1}(k) \partial_k h(k)].$$

Diagonalizing h as $h = S\Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots)$ and the invertible matrix S , the trace can be rewritten as

$$\begin{aligned} \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [S\Lambda^{-1} S^{-1} \partial_k (S\Lambda S^{-1})] \\ &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda + S^{-1} \partial_k S + (\partial_k S^{-1}) S] \\ \text{Tr} [S^{-1} \partial_k S + S \partial_k S^{-1}] &= \text{Tr} [\partial_k (SS^{-1})] = 0, \\ \Rightarrow \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda] = \sum_i \frac{\partial_k \lambda_i}{\lambda_i} = \partial_k \left(\sum_i \ln(\lambda_i) \right) = \partial_k \ln(\det[h]). \end{aligned}$$

Another useful definition of the winding number is through the so-called Q -matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, \dots, N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma |\psi_j\rangle = |\psi_{-j}\rangle$. The Q -matrix is defined as

$$Q = \sum_{j=1}^N (|\psi_j\rangle \langle \psi_j| - |\psi_{-j}\rangle \langle \psi_{-j}|).$$

As a consequence of the chiral symmetry, the Q -matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned} \Gamma Q \Gamma &= \sum_{j=1}^N (|\psi_{-j}\rangle \langle \psi_{-j}| - |\psi_j\rangle \langle \psi_j|) = -Q \\ \Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix} \end{aligned}$$

The fact that the winding number of the q -matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned} Q^2 = 1 & \Rightarrow qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\ QHQ = H & \Rightarrow qh^\dagger q = h^\dagger \end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned} 0 &= \text{Tr} [q^\dagger (h^\dagger)^{-1} \partial_k (h^\dagger q)] \\ &= \text{Tr} [q q^\dagger (h^\dagger)^{-1} \partial_k h^\dagger + h^\dagger (h^\dagger)^{-1} q^\dagger \partial_k q] \\ \Rightarrow \text{Tr} [q^{-1} \partial_k q] &= -\text{Tr} [(h^\dagger)^{-1} \partial_k h^\dagger] \\ &= \text{Tr} [h^{-1} \partial_k h] \end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of N -component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ |h_{2i}\rangle \end{pmatrix} \quad i \in \{1, \dots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ -|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle \langle h_{2i}| \\ |h_{2i}\rangle \langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow q &= \sum_i |h_{1i}\rangle \langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \text{Tr}[q^\dagger \partial_k q] &= \text{Tr} \left[\sum_{i,j} |h_{2i}\rangle \langle h_{1i}| \partial_k (|h_{1j}\rangle \langle h_{2j}|) \right] \\ &= \sum_{i,j} \text{Tr} [|h_{2i}\rangle \langle h_{1i}| (\partial_k |h_{1j}\rangle) \langle h_{2j}|] + \sum_{i,j} \text{Tr} [|h_{2i}\rangle \langle h_{1i}| |h_{1j}\rangle (\partial_k \langle h_{2j}|)] \\ &= \sum_i \langle h_{1i}| (\partial_k |h_{1i}\rangle) + \sum_i (\partial_k \langle h_{2i}|) |h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\text{BZ}} dk (\partial_k \langle h_{2i}|) |h_{2i}\rangle = - \int_{\text{BZ}} dk \langle h_{2i}| (\partial_k |h_{2i}\rangle)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i} \sum_i \int_{\text{BZ}} dk (\langle h_{1i}| \partial_k |h_{1i}\rangle - \langle h_{2i}| \partial_k |h_{2i}\rangle) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$\begin{aligned}
w &= \frac{1}{\pi i} \sum_i \int_{\text{BZ}} dk \langle \Gamma \psi_i^- | \partial_k | \psi_i^- \rangle \\
&= \frac{1}{\pi i} \sum_i \int_{\text{BZ}} dk \langle \psi_i^+ | \partial_k | \psi_i^- \rangle \\
&= \frac{1}{2\pi i} \sum_i \int_{\text{BZ}} dk (\langle h_{1i} | \partial_k | h_{1i} \rangle - \langle h_{2i} | \partial_k | h_{2i} \rangle)
\end{aligned}$$

The quantity $\sum_i \langle \Gamma \psi_i^- | \partial_k | \psi_i^- \rangle$ can be related to a polarisation [8].

1D analysis of topological states

In order to have a chiral Hamiltonian I will choose the orientation of the dipoles in a way, such that the chirality remains. This will be done for each lattice configuration separately, i.e. the dipoles depend on the lattice geometry.

In order to make the achievement of chirality more feasible I only consider nearest neighbour interactions. When talking of chirality the chirality of the matrix U , defined above, is in mind. Recall that

$$\begin{aligned}
U_{n,m} &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\
U_{n,n} &= \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}
\end{aligned}$$

Chirality implies that all couplings between sites on the same sublattice vanish. The nearest neighbour restriction already assures that couplings between different sites of the same sublattice vanish. Left is thus only the onsite potential due to the Taylor expansion of the dipole-dipole potential. The aim is now to choose the orientation of the dipoles in such a way that the sum in $U_{n,n}$ is equal to zero.

Be aware that so far I didn't write explicitly the dependence of U from the direction of atom movement x, y . Hence $U_{n,m}, U_{n,n}$ are themselves two times two matrixes $(U_{n,m}^{\alpha,\beta})_{\alpha,\beta \in \{x,y\}}$ and the Hamiltonian expressed explicitly with respect to the x, y -phonon modes reads

$$H = \sum_{n,\alpha} \omega a_{n,\alpha}^\dagger a_{n,\alpha} + \frac{2}{m\omega} \sum_{n,m,\alpha,\beta} U_{n,m}^{\alpha,\beta} a_{n,\alpha}^\dagger a_{m,\beta}$$

The terms with $a_{n,\alpha}^\dagger a_{n,\alpha}$ can thus be eliminated by redefining the trap frequency.

$$\omega_\alpha = \omega + \frac{2}{m\omega} U_{n,n}^{\alpha,\alpha}$$

Because on the borders $U_{n,n}^{\alpha,\alpha}$ is different from the bulk values, the trap potential for the atoms on the boundary have to be altered to compensate this shift. As I tested in the simulation the existence of separated energies in the bulk band gaps breaks down when the Hamiltonian on the border is different from the bulk.

The terms coupling the two directions of phonon modes on the same side can not be internalized by another quantity. Here the degrees of freedom related to the dipole orientation are exploited to tune the $U_{n,n}^{x,y}$ values to zero, as mentioned above. Because of the nearest neighbour nature of the consideration only two distances play a role, the intracell distance $\mathbf{r}_I$ and the intercell distance $\mathbf{r}_O$.

$$U_{n,n}^{x,y} = \frac{\partial^2 V(\mathbf{r}_I)}{\partial r_{I,x} \partial r_{I,y}} + \frac{\partial^2 V(\mathbf{r}_O)}{\partial r_{O,x} \partial r_{O,y}}$$

The dipoles are then chosen in such a way that this sum vanishes. For open boundary conditions the values of $U_{n,n}^{\alpha,\beta}$ at the borders are different from the bulk values. In the case of $\alpha = \beta$ the difference can be compensated by changed trap frequencies at the boundary. For the xy coupling this is not possible and the interaction due to both neighboring atoms has to be made zero individually such that x - and

y -modes are completely decoupled from each other.

Informed by simulations where the deviation of the border terms in the Hamiltonian has been kept the topologically non-trivial phase breaks down. It might be at first counter intuitive, that reducing the effect of an open border is detrimental to the observation of edge states. It makes, however perfect sense, when remembering that those boundary terms in our case destroy the chiral symmetry. It seems to be necessary for the bulk-boundary condition to hold that not only the momentum space Hamiltonian with periodic boundary conditions but also for the real space Hamiltonian with open boundary conditions has to satisfy chiral symmetry. Otherwise one could choose the Hamiltonian describing the border in a completely arbitrary manner and still observe edge states, which would be quite a strong statement.

Aligned dipoles in plane

I consider a geometry with two sublattices repeating itself in the x -axis and separated by the distance h from each other into the y -axis. Different configurations of this geometry are realized by moving the sublattices against each other in the x -direction. I consider an even amount of lattice, with N atoms in each sublattice. The n th atom of each sublattice A and B takes the position.

$$\begin{aligned}\mathbf{r}_n^A &= n \cdot a \mathbf{e}_x \\ \mathbf{r}_n^B &= n \cdot a \mathbf{e}_x + b \mathbf{e}_x + h \mathbf{e}_y,\end{aligned}$$

where a is the lattice constant, and the parameters b and h can be chosen to realize different lattice configurations. b and h have to be chosen in a way, such that the nearest neighbor restriction makes sense. For any of these geometries there exists a unique orientation of the dipoles which makes the Hamiltonian with periodic boundary conditions chiral.

By contemplating the spectrum in dependence of the parameter b we find energy state that are well separated from the bulk energy Fig. 2. But the separated energy values converge to a non-zero value, when the number of atoms goes to infinity. But this would be necessary for proper topological edge states. I assume this discrepancy occurs because, as I constructed the dipole moment, the components of the Hamiltonian related to the border are non vanishing. Only the bulk Hamiltonian obeys chiral symmetry. It makes of course sense that the chirality is not only required for the momentum space Hamiltonian with periodic boundary conditions but also for the real space Hamiltonian. In order for the bulk boundary correspondence to hold it seems to be necessary that the real space Hamiltonian satisfies the same symmetries as the momentum space Hamiltonian. Seems reasonable.

Fortunately it is still possible to set

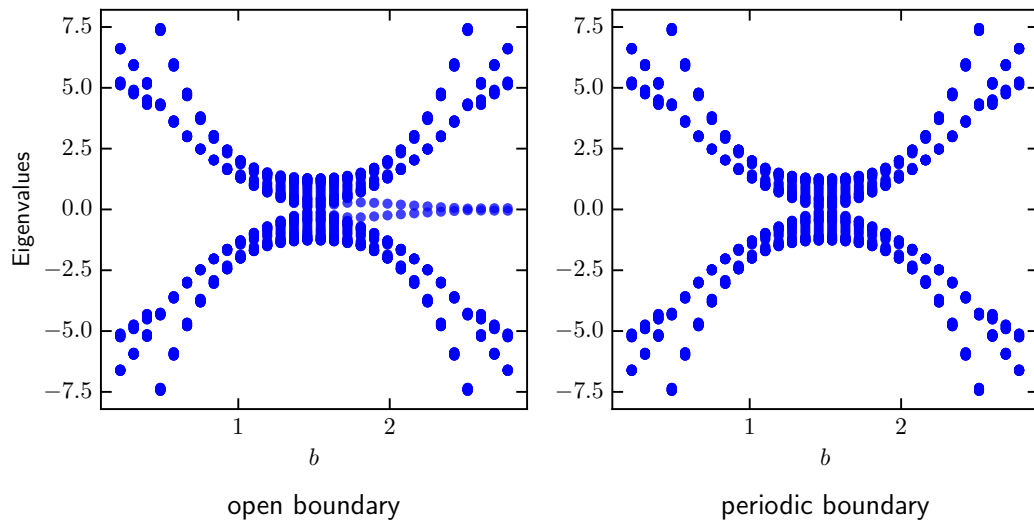


Figure 2: **Spectrum of almost chiral-symmetric Hamiltonian.**

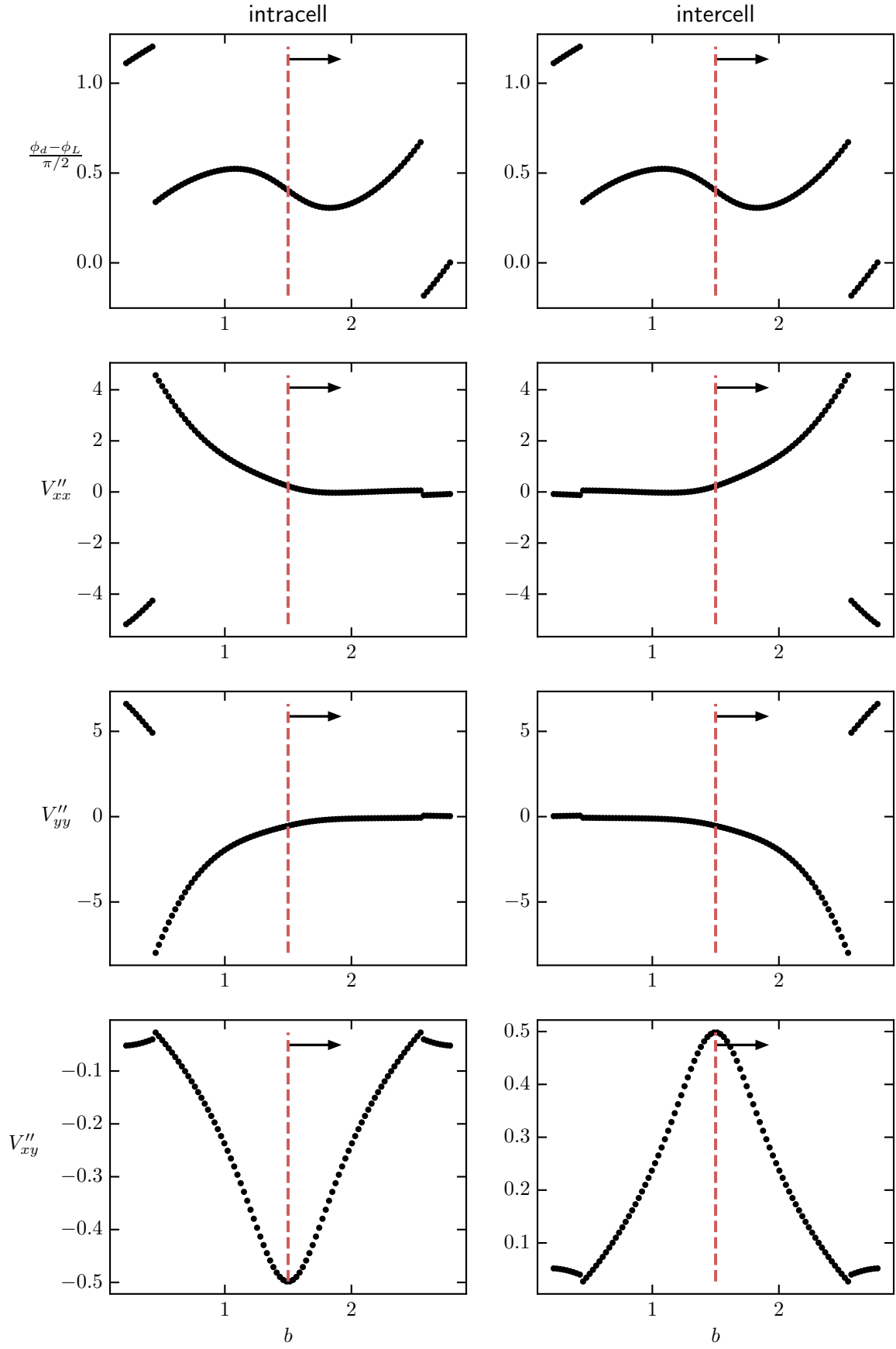
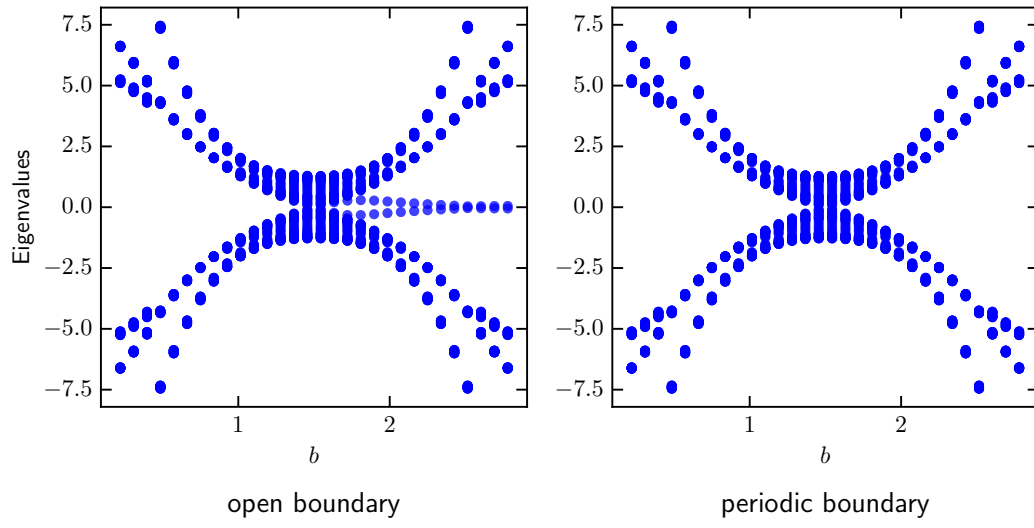
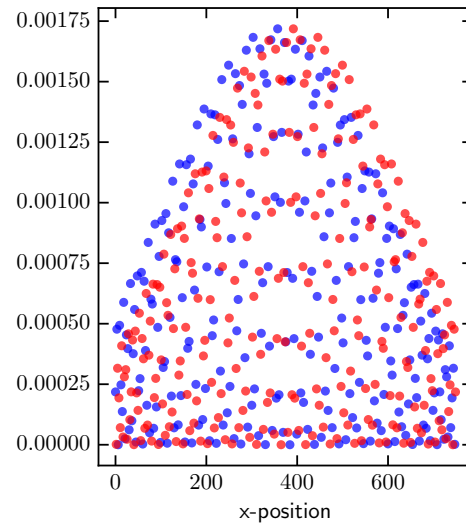
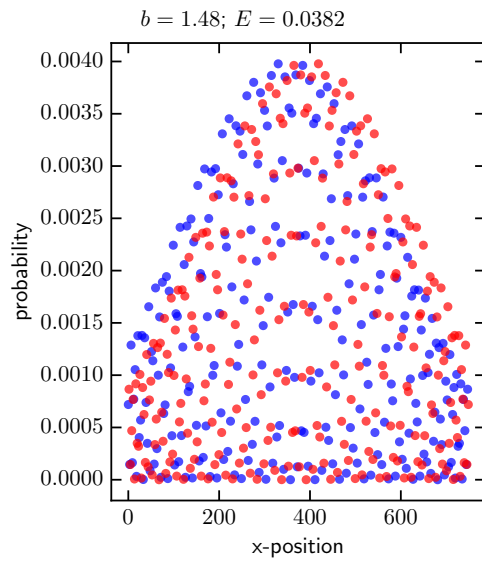
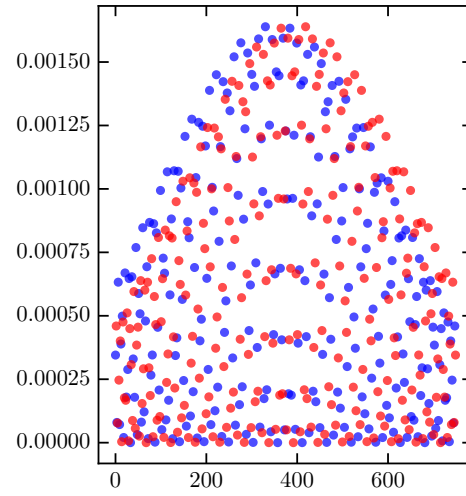
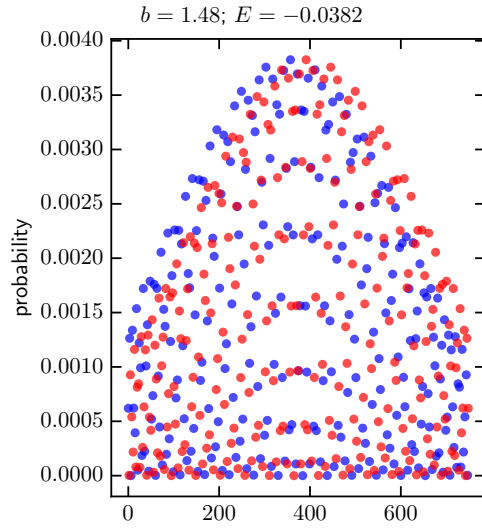


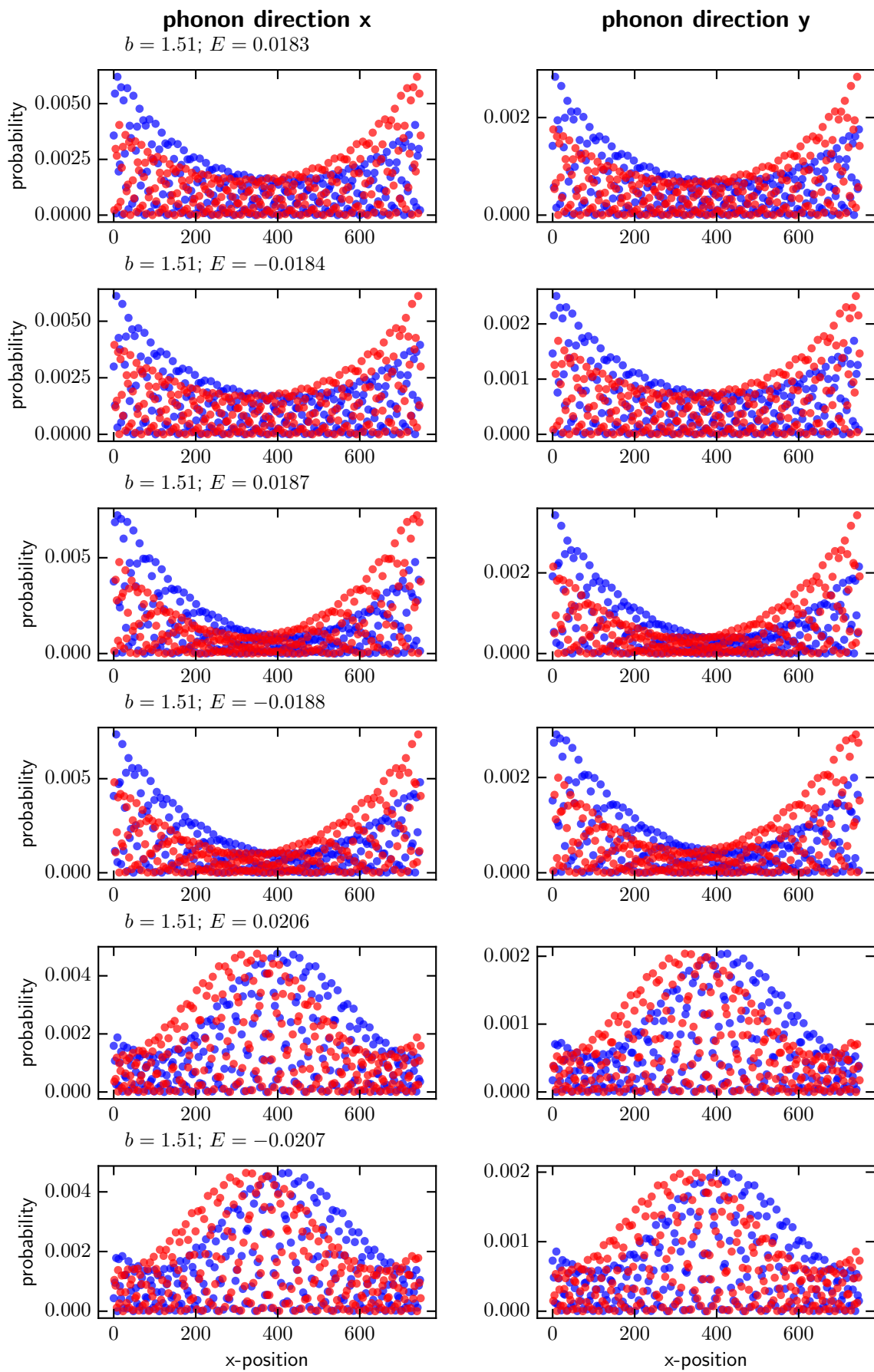
Figure 3: The diagram was created with the parameters $a = 3$, $h = 1.1$. Right to the red dashed line topological phonon states can be found and the winding number is 2.

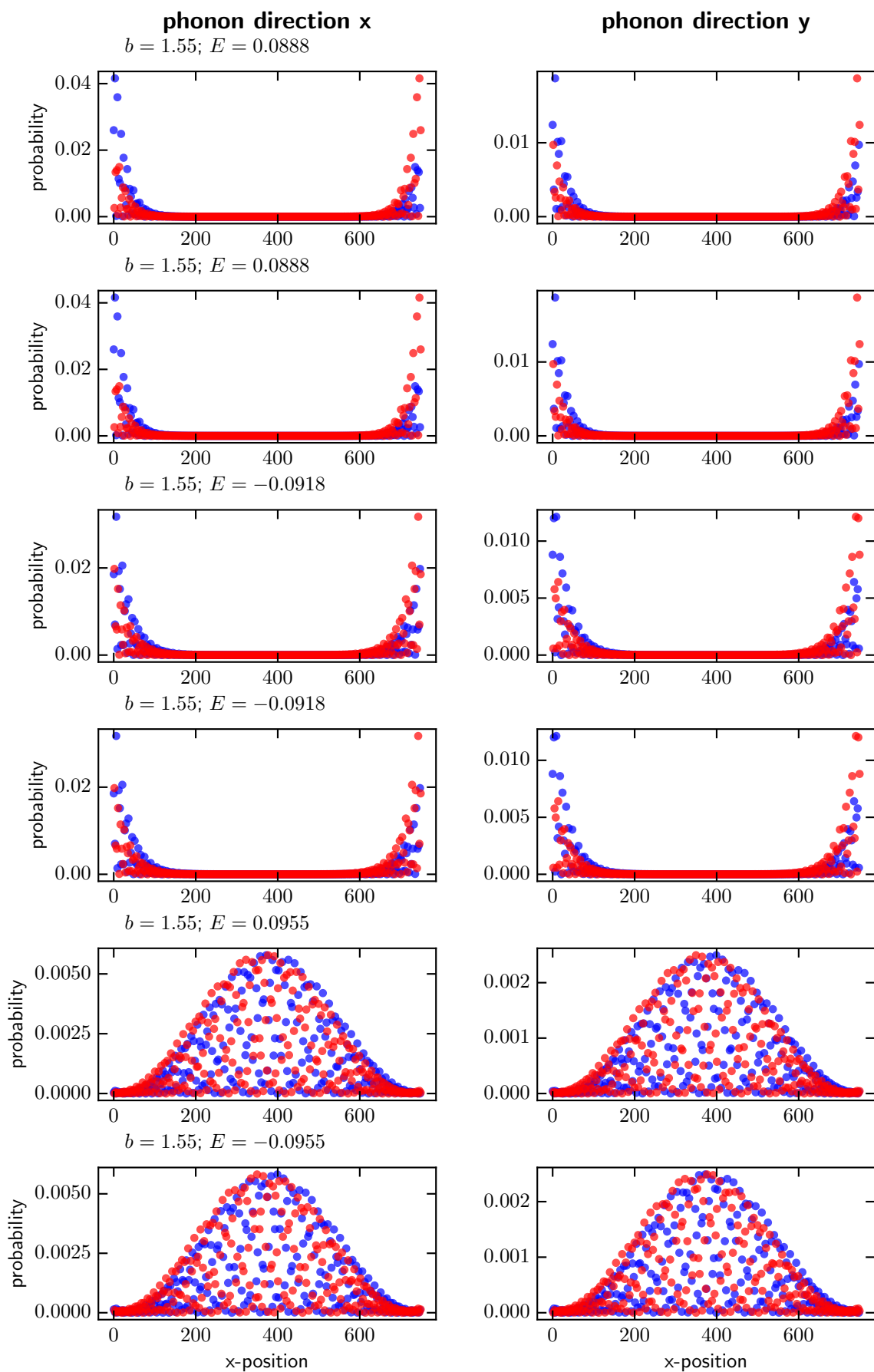


phonon direction x

phonon direction y







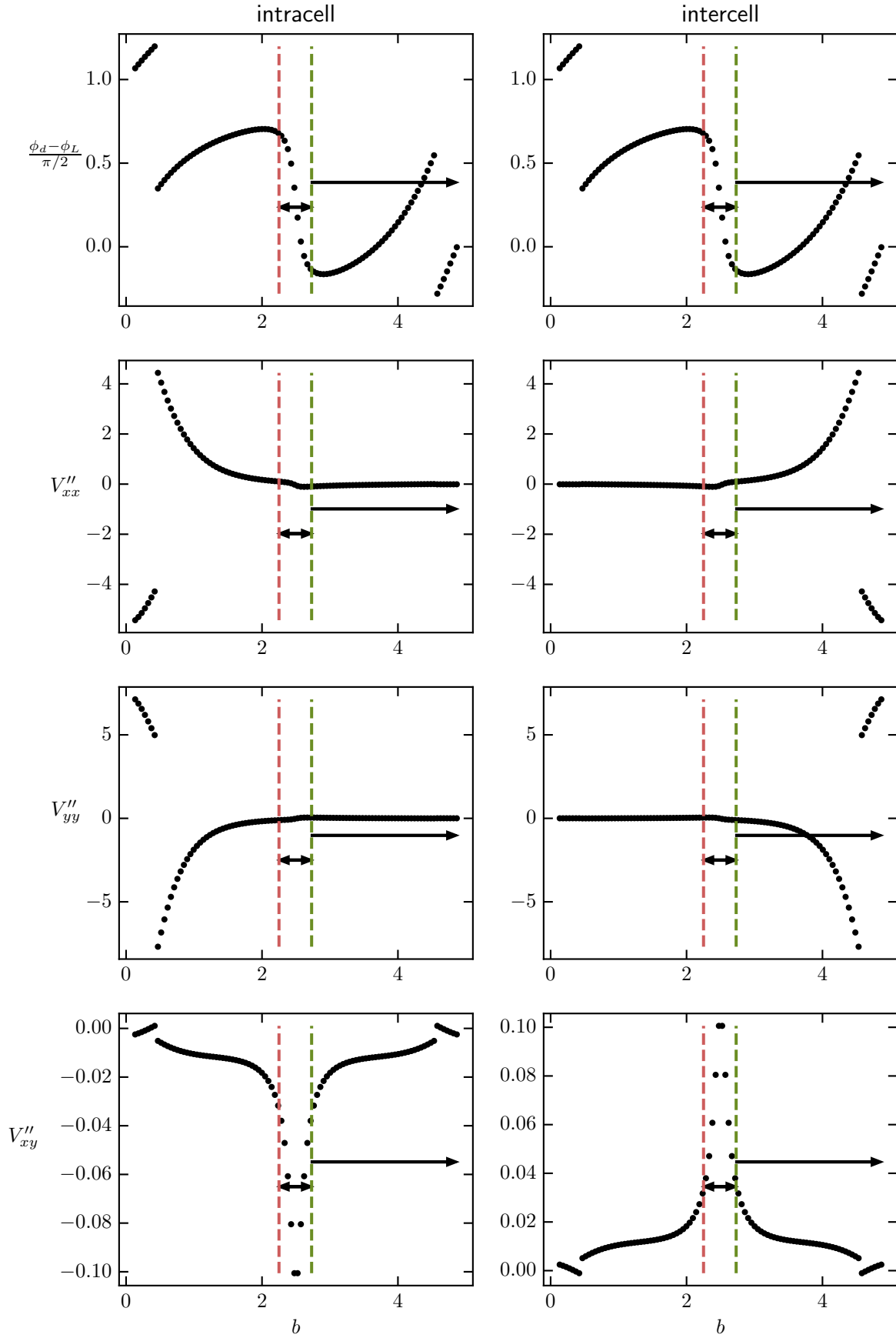
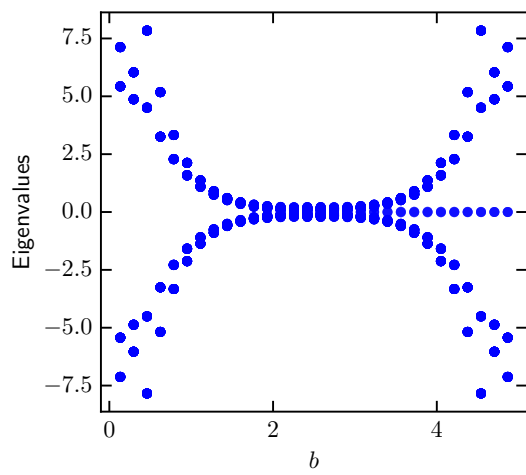
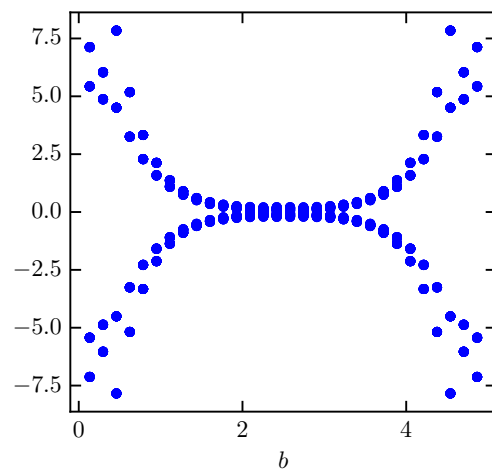


Figure 4: The diagram was created with the parameters $a = 5$, $h = 1.1$. Between the red and the olive dashed line, i.e. between b -values of 2.25-2.73 the winding number is 1, whereas right to the olive dashed line the winding number is 2.

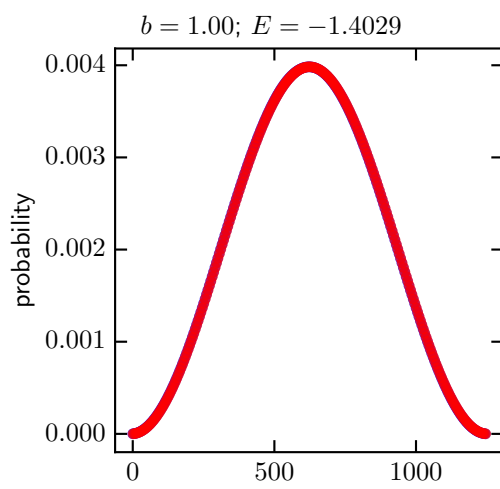


open boundary

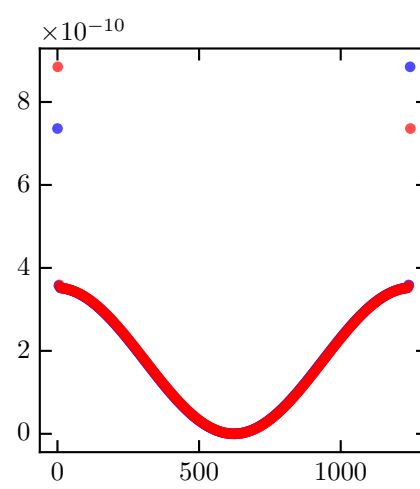


periodic boundary

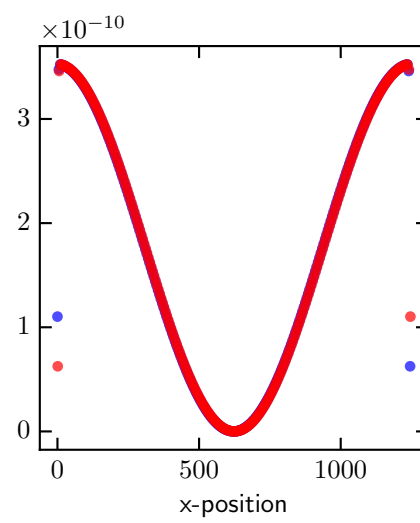
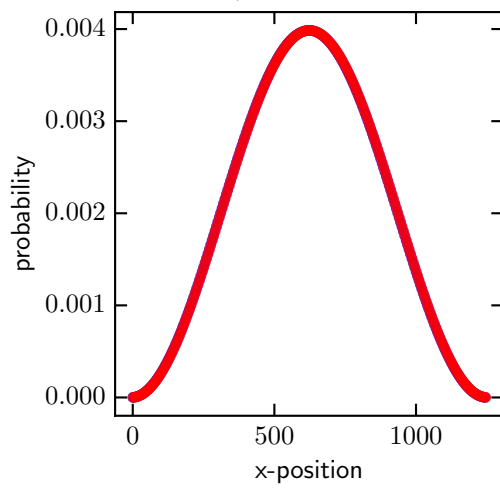
phonon direction x

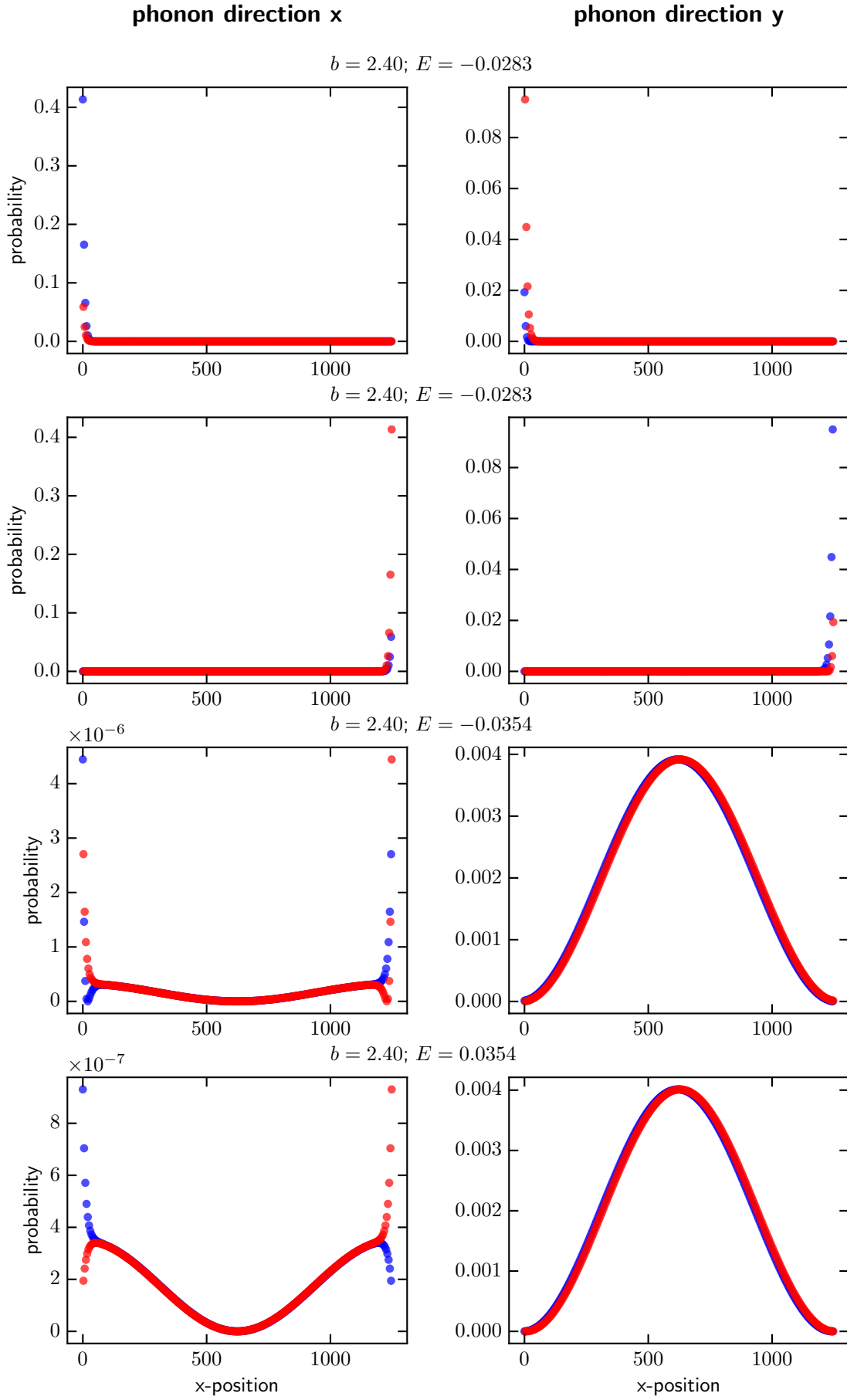


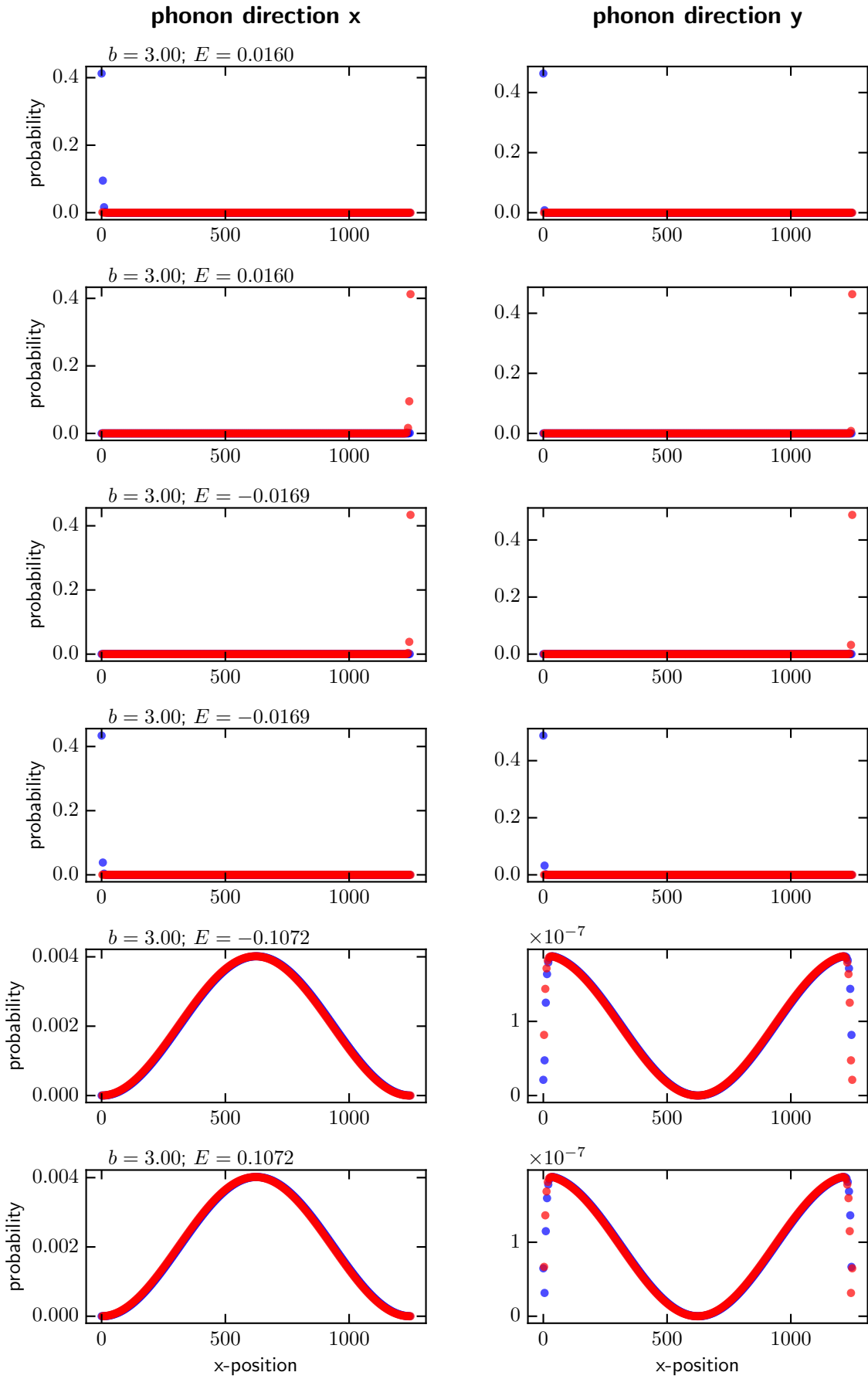
phonon direction y



$b = 1.00; E = 1.4029$







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